

(12) **United States Patent**
Luan et al.

(10) **Patent No.:** **US 12,385,761 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **MULTI-DIMENSIONAL PRECISION SPOT INSPECTION METHOD AND SYSTEM FOR THERMAL POWER EQUIPMENT**

(71) Applicants: **Huaneng Jinan Huangtai Power Generation Co., Ltd.**, Shandong (CN); **Shandong Huaneng Power Generation Co., Ltd.**, Shandong (CN)

(72) Inventors: **Jun Luan**, Shandong (CN); **Xianchao Lu**, Shandong (CN); **Jie Li**, Shandong (CN); **Yuling Wang**, Shandong (CN); **Zhongyu Tian**, Shandong (CN); **Rui Wan**, Shandong (CN); **Kebing Wang**, Shandong (CN); **Jianyong Ren**, Shandong (CN); **Peng Sun**, Shandong (CN); **Shuwen Leng**, Shandong (CN); **Xueli Li**, Shandong (CN)

(73) Assignees: **Huaneng Jinan Huangtai Power Generation Co., Ltd.**, Jinan (CN); **Shandong Huaneng Power Generation Co. Ltd.**, Jinan (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 200 days.

(21) Appl. No.: **18/348,350**

(22) Filed: **Jul. 7, 2023**

(65) **Prior Publication Data**
US 2024/0280378 A1 Aug. 22, 2024

Related U.S. Application Data
(63) Continuation of application No. PCT/CN2023/096717, filed on May 29, 2023.

(30) **Foreign Application Priority Data**
Feb. 20, 2023 (CN) 202310135549.0

(51) **Int. Cl.**
G01D 3/06 (2006.01)
G01D 21/02 (2006.01)

(52) **U.S. Cl.**
CPC **G01D 3/063** (2013.01); **G01D 21/02** (2013.01)

(58) **Field of Classification Search**
CPC G01D 3/063; G01D 21/02; G05B 23/02
See application file for complete search history.

(56) **References Cited**
U.S. PATENT DOCUMENTS

5,844,410 A * 12/1998 Ikuta G01R 15/247
324/96
2015/0355274 A1 * 12/2015 Ross G01R 31/311
324/754.23

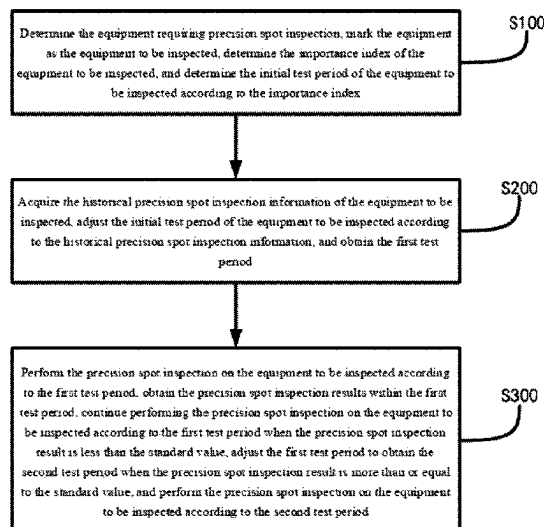
(Continued)

Primary Examiner — Lee E Rodak
Assistant Examiner — Zannatul Ferdous

(57) **ABSTRACT**

A multi-dimensional precision spot inspection method and system for thermal power equipment, and the method includes: determine the equipment requiring precision spot inspection and the initial test period of the equipment to be inspected; acquire the historical precision spot inspection information of the equipment to be inspected, adjust the initial test period of the equipment to be inspected according to the historical precision spot inspection information, and obtain the first test period; perform the precise spot inspection on the equipment to be inspected according to the first test period, obtain the precision spot inspection result within the first test period, and adjust the first test period to obtain the second test period when the precision spot inspection result is more than or equal to the standard value. In the invention, the test period of the thermal power equipment is adjusted, effectively improving the effectiveness and accuracy of the test period.

5 Claims, 2 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2021/0148879 A1* 5/2021 Liu G01N 33/007
2024/0280378 A1* 8/2024 Luan G05B 23/02

* cited by examiner

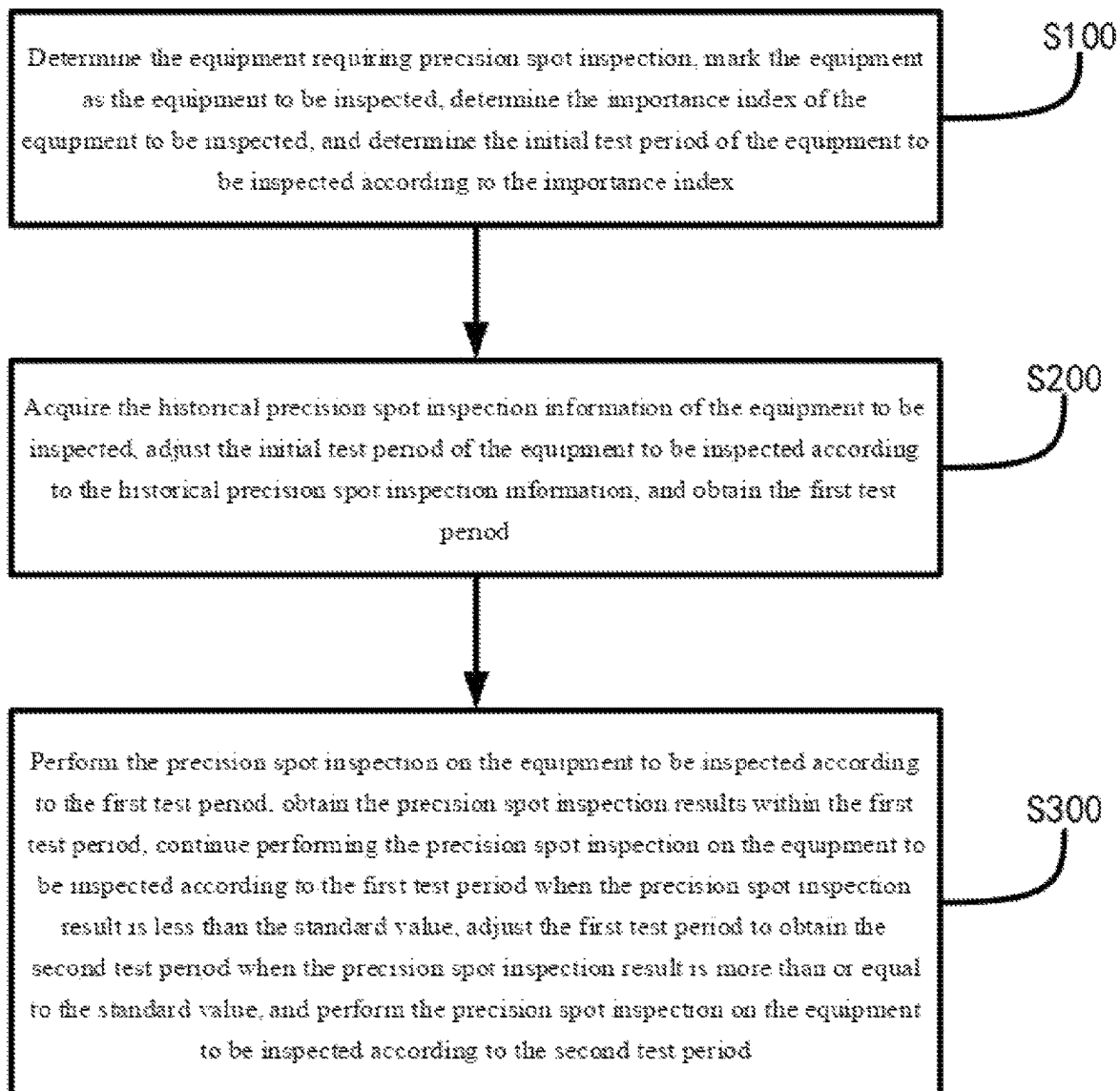


FIG. 1

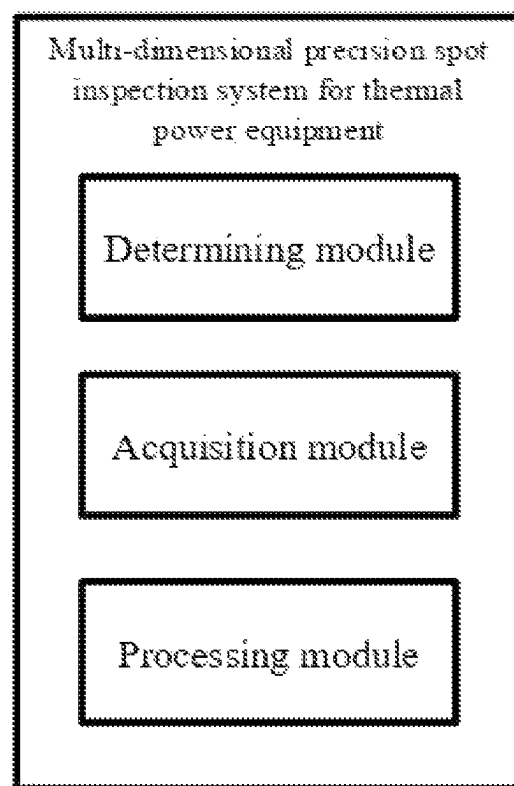


FIG. 2

MULTI-DIMENSIONAL PRECISION SPOT INSPECTION METHOD AND SYSTEM FOR THERMAL POWER EQUIPMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of PCT/CN2023/096717, filed May 29, 2023 and claims priority of Chinese Patent Application No. 202310135549.0, filed on Feb. 20, 2023, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The invention relates to the technical field of thermal power equipment, in particular to a multi-dimensional precision spot inspection method and system for thermal power equipment.

BACKGROUND ART

Precision spot inspection refers to comprehensive equipment testing and investigation by use of precision testing instruments, or determination of vibration, temperature, crack, insulation, temperature rise, deformation and other physical quantities of equipment by use of diagnostic technology, special instruments, tools or special methods without equipment disassembly, and quantitative determination of technical conditions and deterioration degree of equipment by analysis, comparison, determination according to the standards.

As there are numerous thermal power equipment, the thermal power equipment is tested offline by use of precision testing instruments and meters during precision spot inspection. In the normal operation, the physical quantities that characterize the operating state of the equipment are determined by using the vibration, infrared, ultrasonic, current spectrum, oil and other testing instruments, and the technical conditions and deterioration degree of the equipment are determined according to the standards.

The Chinese patent application CN113405822A relates to a precision spot inspection method for thermal power equipment, which includes the following steps: S1, determine the equipment incorporated in the precision spot inspection; S2, determine the location and number of measuring points of the equipment; S3, determine the test period; S4, create a matched technical means according to the working principle and physical and chemical characteristics of each equipment; S5, test by the corresponding technical means according to the test period; S6, evaluate the test results. The test period of the equipment is determined according to the importance of the equipment and dynamically adjusted according to the operating state of the equipment. When the operating state of the equipment deteriorates, the test frequency is increased. The invention has the advantage of dynamically adjusting the test period according to the importance of the equipment and the operating state.

However, how to acquire multi-dimensional information of thermal power equipment for precise spot inspection to timely adjust the test period of thermal power equipment for precision spot inspection has become an urgent problem to be solved.

Content of Invention

In view of this, the invention provides a multi-dimensional precision spot inspection method and system for thermal power equipment, aiming at solving the problem of

how to perform the precision spot inspection for the thermal power equipment according to the multi-dimensional information of the thermal power equipment to timely adjust the test period of the thermal power equipment for precision spot inspection.

On the one hand, the invention provides a multi-dimensional precision point inspection method for thermal power equipment, including:

Determine the equipment requiring precision spot inspection, mark the equipment as the equipment to be inspected, determine the importance index of the equipment to be inspected, and determine the initial test period of the equipment to be inspected according to the importance index;

Acquire the historical precision spot inspection information of the equipment to be inspected, adjust the initial test period of the equipment to be inspected according to the historical precision spot inspection information, and obtain the first test period;

Perform the precision spot inspection on the equipment to be inspected according to the first test period, obtain the precision spot inspection results within the first test period, continue performing the precision spot inspection on the equipment to be inspected according to the first test period when the precision spot inspection result is less than the standard value, adjust the first test period to obtain the second test period when the precision spot inspection result is more than or equal to the standard value, and perform the precision spot inspection on the equipment to be inspected according to the second test period.

Further, determining the initial test period of the equipment to be inspected according to the importance index includes:

Determine the importance index A0 of the equipment to be inspected, preset the first preset importance index A1, the second preset importance index A2, the third preset importance index A3 and the fourth preset importance index A4, and $A1 > A2 > A3 > A4$;

Preset the first preset test period T1, the second preset test period T2, the third preset test period T3 and the fourth preset test period T4, and $0 < T1 < T2 < T3 < T4$;

Determine the initial test period of the equipment to be inspected according to the relationship between the importance index A0 and each preset importance index:

When $A0 \geq A1$, select the first preset test period T1 as the initial test period of the equipment to be inspected;

When $A1 > A0 \geq A2$, select the second preset test period T2 as the initial test period of the equipment to be inspected;

When $A2 > A0 \geq A3$, select the third preset test period T3 as the initial test period of the equipment to be inspected;

When $A3 > A0 \geq A4$, select the fourth preset test period T4 as the initial test period of the equipment to be inspected;

After the i th preset test period T_i is selected as the initial test period of the equipment to be inspected, $i=1,2,3,4$, and then the initial test period of the equipment to be inspected is T_i at this time, which is denoted as the initial test period T_i ;

Further, obtaining the historical precision spot inspection information of the equipment to be inspected, adjusting the initial test period of the equipment to be inspected according to the historical precision spot inspection information and obtaining the first test period include:

After determining the initial test period T_i , acquire the historical precision spot inspection information of the equipment to be inspected, which includes the historical precision spot inspection result data and the historical maintenance data of the equipment to be inspected, and adjust the initial

test test period T_i according to the historical precision spot inspection result data and the historical maintenance data; wherein,

After obtaining the historical precision inspection check information, determine whether the historical precision spot inspection result data and the historical maintenance data are obtained:

If only the historical precision spot inspection result data is obtained, determine the difference between two adjacent historical precision spot inspection result data, adjust the initial test period T_i according to the difference between two adjacent historical precision spot inspection result data to obtain the first test period;

If both the historical precision spot inspection result data and the historical maintenance data are obtained, determine the difference between two adjacent historical precision spot inspection result data, obtain the historical maintenance frequency of the equipment to be inspected, adjust the initial test period T_i according to the difference between two adjacent historical precision spot inspection result data, and then correct the adjusted initial test period T_i according to the historical maintenance frequency to obtain the first test period.

Further, adjusting the initial test period T_i based on the difference between two adjacent historical precision spot inspection result data includes:

After obtaining the historical precision spot inspection result data, establish the historical result matrix B according to the historical precision spot inspection result data, set B ($C_1, D_1, E_1, C_2, D_2, E_2, C_3, D_3, E_3, \dots, C_n, D_n, E_n$), where C_1 to C_n are overall vibration values of the equipment to be inspected which are respectively obtained from the 1st precision spot inspection to the nth precision spot inspection, D_1 to D_n are infrared temperature values of the equipment to be inspected which are respectively obtained from the 1st precision spot inspection to the nth precision spot inspection, E_1 to E_n are ultrasonic values of the equipment to be inspected which are respectively obtained from the 1st precision spot inspection to the nth precision spot inspection;

Preset the preset overall vibration difference c_0 , the preset standard infrared temperature difference d_0 and the preset standard ultrasonic difference e_0 ; preset the first preset adjustment coefficient z_1 , the second preset adjustment coefficient z_2 and the third preset adjustment coefficient z_3 , and $1 > z_1 > z_2 > z_3 > 0.8$;

Obtain the overall vibration values, the infrared temperature values and the ultrasonic values within the last two periods in the historical result matrix B, obtain two adjacent overall vibration differences, denoted as $C_{n-1}-C_n$, obtain two adjacent infrared temperature differences, denoted as $D_{n-1}-D_n$, and obtain two adjacent ultrasonic differences, denoted as $E_{n-1}-E_n$;

Compare $C_{n-1}-C_n$, $D_{n-1}-D_n$ and $E_{n-1}-E_n$ with c_0 , d_0 and e_0 respectively, and select the adjustment coefficient according to the comparison results to adjust the initial test period T_i :

When $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n \leq e_0$, the initial test period T_i is not adjusted;

When $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n \leq e_0$, or $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n \leq e_0$, or $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n > e_0$, select the first preset adjustment coefficient z_1 to adjust the initial test period T_i , the adjusted initial test period is $T_i * z_1$, and $T_i * z_1$ is an integer;

When $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n \leq e_0$, or $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n > e_0$, or $C_{n-1}-$

$C_n > c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n > e_0$, select the second preset adjustment coefficient z_2 to adjust the initial test period T_i , the adjusted initial test period is $T_i * z_2$, and $T_i * z_2$ is an integer;

When $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n > e_0$, select the third preset adjustment coefficient z_3 to adjust the initial test period T_i , the adjusted initial test period is $T_i * z_3$, and $T_i * z_3$ is an integer;

After the mth preset adjustment coefficient z_m is selected to adjust the initial test period T_i , $m=1,2,3$, and then the adjusted initial test period $T_i * z_m$ is taken as the first test period.

Further, taking the adjusted initial test period $T_i * z_m$ as the first test period includes:

After establishing the historical result matrix B based on the historical precision spot inspection result data, determine the overall vibration average c_{01} , the infrared temperature average d_{01} and the ultrasonic average e_{01} according to the historical result matrix B, where

$$c_{01} = (C_1 + C_2 + C_3 + \dots + C_n)/n, d_{01} =$$

$$(D_1 + D_2 + D_3 + \dots + D_n)/n, \text{ and } e_{01} = (E_1 + E_2 + E_3 + \dots + E_n)/n;$$

Preset the preset standard overall vibration average c_a , the preset standard infrared temperature average d_a and the preset standard ultrasonic average e_a , and adjust the adjusted initial test period $T_i * z_m$ again according to the relationship between the overall vibration average c_{01} , the infrared temperature average d_{01} and the ultrasonic average e_{01} , and the preset standard overall vibration average c_a , the preset standard infrared temperature average d_a and the preset standard ultrasonic average e_a respectively:

When $c_{01} \leq c_a$, $d_{01} \leq d_a$ and $e_{01} \leq e_a$, the adjusted initial test period $T_i * z_m$ is not adjusted again.

When $c_{01} > c_a$, $d_{01} \leq d_a$ and $e_{01} \leq e_a$, or $c_{01} \leq c_a$, $d_{01} > d_a$ and $e_{01} \leq e_a$, or $c_{01} \leq c_a$, $d_{01} \leq d_a$ and $e_{01} > e_a$, select the first preset adjustment coefficient z_1 to adjust the adjusted initial test period $T_i * z_m$ again, the readjusted test period is $T_i * z_m * z_1$, and $T_i * z_m * z_1$ is an integer;

When $c_{01} > c_a$, $d_{01} > d_a$ and $e_{01} \leq e_a$, or $c_{01} \leq c_a$, $d_{01} > d_a$ and $e_{01} \leq e_a$, or $c_{01} > c_a$, $d_{01} \leq d_a$ and $e_{01} > e_a$, select the second preset adjustment coefficient z_2 to adjust the adjusted initial test period $T_i * z_m$ again, the readjusted test period is $T_i * z_m * z_2$, and $T_i * z_m * z_2$ is an integer;

When $c_{01} > c_a$, $d_{01} > d_a$ and $e_{01} > e_a$, select the third preset adjustment coefficient z_3 to adjust the adjusted initial test period $T_i * z_m$ again, the readjusted test period is $T_i * z_m * z_3$, and $T_i * z_m * z_3$ is an integer.

After the mth preset adjustment coefficient z_m is selected to adjust the adjusted initial test period $T_i * z_m$ again, $m=1, 2, 3$, and then the adjusted initial test period $T_i * z_m * z_m$ is taken as the first test period.

Further, adjusting the initial test period T_i according to the difference between two adjacent historical precision spot inspection result data and then correcting the adjusted initial test period T_i according to the historical maintenance frequency to obtain the first test period include:

After obtaining the historical maintenance data, determine the historical maintenance frequency Q_0 of the equipment to be inspected according to the historical maintenance data;

Preset the first preset maintenance frequency Q_1 , the second preset maintenance frequency Q_2 , the third preset maintenance frequency Q_3 and the fourth preset maintenance frequency Q_4 , and $Q_1 < Q_2 < Q_3 < Q_4$; preset the first

5

preset correction coefficient y_1 , the second preset correction coefficient y_2 , the third preset correction coefficient y_3 and the fourth preset correction coefficient y_4 , and $1 < y_1 < y_2 < y_3 < y_4 < 1.5$;

After obtaining the adjusted initial test period Ti^*zm or Ti^*zm^*zm , select the correction coefficient according to the relationship between the historical maintenance frequency Q_0 and each preset maintenance frequency, and correct the adjusted initial test period Ti^*zm or Ti^*zm^*zm :

When $Q_1 \leq Q_0 < Q_2$, select the first preset correction coefficient y_1 to correct the adjusted initial test period Ti^*zm or Ti^*zm^*zm , and the corrected test period is $Ti^*zm^*y_1$ or $Ti^*zm^*zm^*y_1$;

When $Q_2 \leq Q_0 < Q_3$, select the second preset correction coefficient y_2 to correct the adjusted initial test period Ti^*zm or Ti^*zm^*zm , and the corrected test period is $Ti^*zm^*y_2$ or $Ti^*zm^*zm^*y_2$;

When $Q_3 \leq Q_0 < Q_4$, select the third preset correction coefficient y_3 to correct the adjusted initial test period Ti^*zm or Ti^*zm^*zm , and the corrected test period is $Ti^*zm^*y_3$ or $Ti^*zm^*zm^*y_3$;

When $Q_4 \leq Q_0$, select the fourth preset correction coefficient y_4 to correct the adjusted initial test period Ti^*zm or Ti^*zm^*zm , and the corrected test period is $Ti^*zm^*y_4$ or $Ti^*zm^*zm^*y_4$;

After selecting the i th preset correction coefficient y_i to correct the adjusted initial test period Ti^*zm or Ti^*zm^*zm , $i=1,2,3,4$, and the corrected test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$ is taken as the first test period.

Further, taking the corrected test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$ as the first test period includes:

Perform the precision spot inspection on the equipment to be inspected according to the first test period, and obtain the precision spot inspection results within the first test period, which includes the overall vibration value ΔC_1 , the infrared temperature value ΔD_1 and the ultrasonic value ΔE_1 within the first test period; wherein,

When the overall vibration value ΔC_1 , the infrared temperature value ΔD_1 and the ultrasonic value ΔE_1 within the first test period are less than the standard values, do not adjust the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$, and continue to perform the precision spot inspection on the equipment to be inspected according to the first test period;

When at least one of the overall vibration value ΔC_1 , the infrared temperature value ΔD_1 and the ultrasonic value ΔE_1 within the first test period is more than or equal to the standard value, adjust the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$ to obtain the second test period, and perform the precision spot inspection on the equipment to be inspected according to the second test period.

Further, presetting the standard values which include the preset standard overall vibration value C_{10} , the preset standard infrared temperature value D_{10} and the preset standard ultrasonic value E_{10} and obtaining the overall vibration value ΔC_1 , the infrared temperature value ΔD_1 and the ultrasonic value ΔE_1 within the first test period include:

Preset the first compensation coefficient r_1 , the second compensation coefficient r_2 , the third compensation coefficient r_3 and the fourth compensation coefficient r_4 , and $0.8 < r_1 < r_2 < r_3 < r_4 < 1$;

When $\Delta C_1 < C_{10}$, $\Delta D_1 < D_{10}$ and $\Delta E_1 < E_{10}$, the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$ is not adjusted;

When at least one of the overall vibration value ΔC_1 , the infrared temperature value ΔD_1 and the ultrasonic value

6

ΔE_1 within the first test period is more than or equal to the standard value, adjust the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$, specifically:

When $\Delta C_1 \geq C_{10}$, $\Delta D_1 < D_{10}$ and $\Delta E_1 < E_{10}$, or $\Delta C_1 < C_{10}$, $\Delta D_1 \geq D_{10}$ and $\Delta E_1 < E_{10}$, or $\Delta C_1 < C_{10}$, $\Delta D_1 < D_{10}$ and $\Delta E_1 \geq E_{10}$, select the first compensation coefficient r_1 to compensate the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$, and the compensated test period is $Ti^*zm^*y_i^*r_1$ or $Ti^*zm^*zm^*y_i^*r_1$, which is taken as the second test period;

When $\Delta C_1 \geq C_{10}$, $\Delta D_1 \geq D_{10}$ and $\Delta E_1 < E_{10}$, or $\Delta C_1 \geq C_{10}$, $\Delta D_1 < D_{10}$ and $\Delta E_1 \geq E_{10}$, or $\Delta C_1 < C_{10}$, $\Delta D_1 \geq D_{10}$ and $\Delta E_1 \geq E_{10}$, select the second compensation coefficient r_2 to compensate the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$, and the compensated test period is $Ti^*zm^*y_i^*r_2$ or $Ti^*zm^*zm^*y_i^*r_2$, which is taken as the second test period;

When $\Delta C_1 \geq C_{10}$, $\Delta D_1 \geq D_{10}$ and $\Delta E_1 \geq E_{10}$, select the third compensation coefficient r_3 to compensate the first test period $Ti^*zm^*y_i$ or $Ti^*zm^*zm^*y_i$, and the compensated test period is $Ti^*zm^*y_i^*r_3$ or $Ti^*zm^*zm^*y_i^*r_3$, which is taken as the second test period.

Compared with the prior art, the invention has the beneficial effects of determining the equipment requiring precision spot inspection, determining the importance index of the equipment to be inspected, determining the initial test period of the equipment to be inspected according to the importance index, acquiring the historical precision spot inspection of the equipment to be inspected, adjusting the initial test period of the equipment to be inspected according to the historical precision spot inspection information, obtaining the first test period, performing the precision spot inspection on the equipment to be inspected according to the first test period, obtaining the precision spot inspection results within the first test period, continuing performing the precision spot inspection on the equipment to be inspected according to the first test period when the precision spot inspection result is less than the standard value, adjusting the first test period to obtain the second test period when the precision spot inspection result is more than or equal to the standard value, and performing the precision spot inspection on the equipment to be inspected according to the second test period. In the invention, the test period of the thermal power equipment is adjusted by acquiring the multi-dimensional data information of the thermal power equipment, effectively improving the effectiveness and accuracy of the test period.

On the other hand, the invention also provides a multi-dimensional precision point inspection system for thermal power equipment, comprising:

A determining module, which is used to determine the equipment requiring precision spot inspection, mark the equipment as the equipment to be inspected, determine the importance index of the equipment to be inspected, and determine the initial test period of the equipment to be inspected according to the importance index;

An acquisition module, which is used to acquire the historical precision spot inspection information of the equipment to be inspected, adjust the initial test period of the equipment to be inspected according to the historical precision spot inspection information, and obtain the first test period;

A processing module, which is used to perform the precision spot inspection on the equipment to be inspected according to the first test period, obtain the precision spot inspection results within the first test period, continue performing the precision spot inspection on the equipment to be inspected according to the first test period when the precision spot inspection result is less than the standard value,

adjust the first test period to obtain the second test period when the precision spot inspection result is more than or equal to the standard value, and perform the precision spot inspection on the equipment to be inspected according to the second test period.

Further, the determining module is also used to determine the initial test period of the equipment to be inspected according to the importance index, including:

Determine the importance index A0 of the equipment to be inspected, preset the first preset importance index A1, the second preset importance index A2, the third preset importance index A3 and the fourth preset importance index A4, and $A1 > A2 > A3 > A4$;

Preset the first preset test period T1, the second preset test period T2, the third preset test period T3 and the fourth preset test period T4, and $0 \leq T1 < T2 < T3 < T4$;

Determine the initial test period of the equipment to be inspected according to the relationship between the importance index A0 and each preset importance index:

When $A0 \geq A1$, select the first preset test period T1 as the initial test period of the equipment to be inspected;

When $A1 > A0 \geq A2$, select the second preset test period T2 as the initial test period of the equipment to be inspected;

When $A2 > A0 \geq A3$, select the third preset test period T3 as the initial test period of the equipment to be inspected;

When $A3 > A0 \geq A4$, select the fourth preset test period T4 as the initial test period of the equipment to be inspected;

After the *i*th preset test period *T_i* is selected as the initial test period of the equipment to be inspected, $i=1,2,3,4$, and then the initial test period of the equipment to be inspected is *T_i* at this time, which is denoted as the initial test period *T_i*.

It is understandable that the multi-dimensional precision spot inspection method and system for the thermal power equipment have the same beneficial effects and will not be repeated here.

DESCRIPTION OF DRAWINGS

By reading the detailed description of the preferred embodiments below, various other advantages and benefits will become clear to the ordinary technicians in the field. The drawings are intended only for the purpose of showing a preferred embodiment and are not considered to be a limitation to the invention. Moreover, the same parts are represented by the same reference marks throughout the drawings. In the drawings:

FIG. 1 is the flowchart of a multi-dimensional precision spot inspection method for thermal power equipment provided by the embodiment of the invention;

FIG. 2 is the functional block diagram of a multi-dimensional precision spot inspection method for thermal power equipment provided by the embodiment of the invention.

EMBODIMENTS

The exemplary embodiments of the invention are described in more detail below with reference to the drawings. Although the exemplary embodiments of the invention are shown in the drawings, it should be understood that the invention may be implemented in various forms and should not be limited by the embodiments set forth herein. On the contrary, these embodiments are provided to enable a more thorough understanding of the invention and a complete communication of the scope of the invention to the technicians in the field. It should be noted that the embodiments and features in the embodiments of the invention can be

combined with each other without conflict. The invention is described in detail below with reference to the drawings and in conjunction with the embodiments.

As shown in FIG. 1, the embodiment discloses a multi-dimensional precision spot inspection method for thermal power equipment, including the following steps:

Step S100: Determine the equipment requiring precision spot inspection, mark the equipment as the equipment to be inspected, determine the importance index of the equipment to be inspected, and determine the initial test period of the equipment to be inspected according to the importance index.

Specifically, determining the initial test period of the equipment to be inspected according to the importance index includes:

Determine the importance index A0 of the equipment to be inspected, preset the first preset importance index A1, the second preset importance index A2, the third preset importance index A3 and the fourth preset importance index A4, and $A1 > A2 > A3 > A4$;

Preset the first preset test period T1, the second preset test period T2, the third preset test period T3 and the fourth preset test period T4, and $0 \leq T1 < T2 < T3 < T4$;

Determine the initial test period of the equipment to be inspected according to the relationship between the importance index A0 and each preset importance index:

When $A0 \geq A1$, select the first preset test period T1 as the initial test period of the equipment to be inspected;

When $A1 > A0 \geq A2$, select the second preset test period T2 as the initial test period of the equipment to be inspected;

When $A2 > A0 \geq A3$, select the third preset test period T3 as the initial test period of the equipment to be inspected;

When $A3 > A0 \geq A4$, select the fourth preset test period T4 as the initial test period of the equipment to be inspected;

After the *i*th preset test period *T_i* is selected as the initial test period of the equipment to be inspected, $i=1,2,3,4$, and then the initial test period of the equipment to be inspected is *T_i* at this time, which is denoted as the initial test period *T_i*.

Step S200: Acquire the historical precision spot inspection information of the equipment to be inspected, adjust the initial test period of the equipment to be inspected according to the historical precision spot inspection information, and obtain the first test period;

Specifically, obtaining the historical precision spot inspection information of the equipment to be inspected, adjusting the initial test period of the equipment to be inspected according to the historical precision spot inspection information and obtaining the first test period include:

After determining the initial test period *T_i*, acquire the historical precision spot inspection information of the equipment to be inspected, which includes the historical precision spot inspection result data and the historical maintenance data of the equipment to be inspected, and adjust the initial test period *T_i* according to the historical precision spot inspection result data and the historical maintenance data; wherein,

After obtaining the historical precision inspection check information, determine whether the historical precision spot inspection result data and the historical maintenance data are obtained:

If only the historical precision spot inspection result data is obtained, determine the difference between two adjacent historical precision spot inspection result data, adjust the initial test period *T_i* according to the difference between two adjacent historical precision spot inspection result data to obtain the first test period;

If both the historical precision spot inspection result data and the historical maintenance data are obtained, determine the difference between two adjacent historical precision spot inspection result data, obtain the historical maintenance frequency of the equipment to be inspected, adjust the initial test period T_i according to the difference between two adjacent historical precision spot inspection result data, and then correct the adjusted initial test period T_i according to the historical maintenance frequency to obtain the first test period.

Specifically, adjusting the initial test period T_i based on the difference between two adjacent historical precision spot inspection result data includes:

After obtaining the historical precision spot inspection result data, establish the historical result matrix B according to the historical precision spot inspection result data, set B ($C_1, D_1, E_1, C_2, D_2, E_2, C_3, D_3, E_3, \dots, C_n, D_n, E_n$), where C_1 to C_n are overall vibration values of the equipment to be inspected which are respectively obtained from the 1st precision spot inspection to the nth precision spot inspection, D_1 to D_n are infrared temperature values of the equipment to be inspected which are respectively obtained from the 1st precision spot inspection to the nth precision spot inspection, E_1 to E_n are ultrasonic values of the equipment to be inspected which are respectively obtained from the 1st precision spot inspection to the nth precision spot inspection;

Preset the preset overall vibration difference c_0 , the preset standard infrared temperature difference d_0 and the preset standard ultrasonic difference e_0 ; preset the first preset adjustment coefficient z_1 , the second preset adjustment coefficient z_2 and the third preset adjustment coefficient z_3 , and $1 > z_1 > z_2 > z_3 > 0.8$;

Obtain the overall vibration values, the infrared temperature values and the ultrasonic values within the last two periods in the historical result matrix B, obtain two adjacent overall vibration differences, denoted as $C_{n-1}-C_n$, obtain two adjacent infrared temperature differences, denoted as $D_{n-1}-D_n$, and obtain two adjacent ultrasonic differences, denoted as $E_{n-1}-E_n$;

Compare $C_{n-1}-C_n$, $D_{n-1}-D_n$ and $E_{n-1}-E_n$ with c_0 , d_0 and e_0 respectively, and select the adjustment coefficient according to the comparison results to adjust the initial test period T_i :

When $C_{n-1}-C_n < c_0$, $D_{n-1}-D_n < d_0$ and $E_{n-1}-E_n < e_0$, the initial test period T_i is not adjusted;

When $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n \leq e_0$, or $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n \leq e_0$, or $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n > e_0$, select the first preset adjustment coefficient z_1 to adjust the initial test period T_i , the adjusted initial test period is $T_i * z_1$, and $T_i * z_1$ is an integer;

When $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n \leq e_0$, or $C_{n-1}-C_n \leq c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n > e_0$, or $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n \leq d_0$ and $E_{n-1}-E_n > e_0$, select the second preset adjustment coefficient z_2 to adjust the initial test period T_i , the adjusted initial test period is $T_i * z_2$, and $T_i * z_2$ is an integer;

When $C_{n-1}-C_n > c_0$, $D_{n-1}-D_n > d_0$ and $E_{n-1}-E_n > e_0$, select the third preset adjustment coefficient z_3 to adjust the initial test period T_i , the adjusted initial test period is $T_i * z_3$, and $T_i * z_3$ is an integer;

After the mth preset adjustment coefficient z_m is selected to adjust the initial test period T_i , $m=1,2,3$, and then the adjusted initial test period $T_i * z_m$ is taken as the first test period.

Specifically, taking the adjusted initial test period $T_i * z_m$ as the first test period includes:

After establishing the historical result matrix B based on the historical precision spot inspection result data, determine the overall vibration average c_01 , the infrared temperature average d_01 and the ultrasonic average e_01 according to the historical result matrix B, where $c_01 = (C_1 + C_2 + C_3 + \dots + C_n)/n$, $d_01 = (D_1 + D_2 + D_3 + \dots + D_n)/n$, and $e_01 = (E_1 + E_2 + E_3 + \dots + E_n)/n$;

Preset the preset standard overall vibration average c_a , the preset standard infrared temperature average d_a and the preset standard ultrasonic average e_a , and adjust the adjusted initial test period $T_i * z_m$ again according to the relationship between the overall vibration average c_01 , the infrared temperature average d_01 and the ultrasonic average e_01 , and the preset standard overall vibration average c_a , the preset standard infrared temperature average d_a and the preset standard ultrasonic average e_a respectively;

When $c_01 \leq c_a$, $d_01 \leq d_a$ and $e_01 \leq e_a$, the adjusted initial test period $T_i * z_m$ is not adjusted again.

When $c_01 > c_a$, $d_01 \leq d_a$ and $e_01 \leq e_a$, or $c_01 \leq c_a$, $d_01 > d_a$ and $e_01 \leq e_a$, or $c_01 \leq c_a$, $d_01 \leq d_a$ and $e_01 > e_a$, select the first preset adjustment coefficient z_1 to adjust the adjusted initial test period $T_i * z_m$ again, the readjusted test period is $T_i * z_m * z_1$, and $T_i * z_m * z_1$ is an integer;

When $c_01 > c_a$, $d_01 > d_a$ and $e_01 \leq e_a$, or $c_01 \leq c_a$, $d_01 > d_a$ and $e_01 \leq e_a$, or $c_01 > c_a$, $d_01 \leq d_a$ and $e_01 > e_a$, select the second preset adjustment coefficient z_2 to adjust the adjusted initial test period $T_i * z_m$ again, the readjusted test period is $T_i * z_m * z_2$, and $T_i * z_m * z_2$ is an integer;

When $c_01 > c_a$, $d_01 > d_a$ and $e_01 > e_a$, select the third preset adjustment coefficient z_3 to adjust the adjusted initial test period $T_i * z_m$ again, the readjusted test period is $T_i * z_m * z_3$, and $T_i * z_m * z_3$ is an integer.

After the mth preset adjustment coefficient z_m is selected to adjust the adjusted initial test period $T_i * z_m$ again, $m=1, 2, 3$, and then the adjusted initial test period $T_i * z_m * z_m$ is taken as the first test period.

Specifically, adjusting the initial test period T_i according to the difference between two adjacent historical precision spot inspection result data and then correcting the adjusted initial test period T_i according to the historical maintenance frequency to obtain the first test period include:

After obtaining the historical maintenance data, determine the historical maintenance frequency Q_0 of the equipment to be inspected according to the historical maintenance data;

Preset the first preset maintenance frequency Q_1 , the second preset maintenance frequency Q_2 , the third preset maintenance frequency Q_3 and the fourth preset maintenance frequency Q_4 , and $Q_1 < Q_2 < Q_3 < Q_4$; preset the first preset correction coefficient y_1 , the second preset correction coefficient y_2 , the third preset correction coefficient y_3 and the fourth preset correction coefficient y_4 , and $1 < y_1 < y_2 < y_3 < y_4 < 1.5$;

After obtaining the adjusted initial test period $T_i * z_m$ or $T_i * z_m * z_m$, select the correction coefficient according to the relationship between the historical maintenance frequency Q_0 and each preset maintenance frequency, and correct the adjusted initial test period $T_i * z_m$ or $T_i * z_m * z_m$;

When $Q_1 \leq Q_0 \leq Q_2$, select the first preset correction coefficient y_1 to correct the adjusted initial test period $T_i * z_m$ or $T_i * z_m * z_m$, and the corrected test period is $T_i * z_m * y_1$ or $T_i * z_m * z_m * y_1$;

When $Q_2 \leq Q_0 \leq Q_3$, select the second preset correction coefficient y_2 to correct the adjusted initial test period $T_i * z_m$ or $T_i * z_m * z_m$, and the corrected test period is $T_i * z_m * y_2$ or $T_i * z_m * z_m * y_2$;

11

When $Q3 \leq Q0 \leq Q4$, select the third preset correction coefficient $y3$ to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, and the corrected test period is $Ti*zm*y3$ or $Ti*zm*zm*y3$;

When $Q4 \leq Q0$, select the fourth preset correction coefficient $y4$ to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, and the corrected test period is $Ti*zm*y4$ or $Ti*zm*zm*y4$;

After selecting the i th preset correction coefficient yi to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, $i=1,2,3,4$, and the corrected test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ is taken as the first test period.

Step S300: Perform the precision spot inspection on the equipment to be inspected according to the first test period, obtain the precision spot inspection results within the first test period, continue performing the precision spot inspection on the equipment to be inspected according to the first test period when the precision spot inspection result is less than the standard value, adjust the first test period to obtain the second test period when the precision spot inspection result is more than or equal to the standard value, and perform the precision spot inspection on the equipment to be inspected according to the second test period.

Specifically, taking the corrected test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ as the first test period includes:

Perform the precision spot inspection on the equipment to be inspected according to the first test period, and obtain the precision spot inspection results within the first test period, which includes the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period; wherein,

When the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period are less than the standard values, do not adjust the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and continue to perform the precision spot inspection on the equipment to be inspected according to the first test period;

When at least one of the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period is more than or equal to the standard value, adjust the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ to obtain the second test period, and perform the precision spot inspection on the equipment to be inspected according to the second test period.

Specifically, presetting the standard values which include the preset standard overall vibration value $C10$, the preset standard infrared temperature value $D10$ and the preset standard ultrasonic value $E10$ and obtaining the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period include:

Preset the first compensation coefficient $r1$, the second compensation coefficient $r2$, the third compensation coefficient $r3$ and the fourth compensation coefficient $r4$, and $0.8 < r1 < r2 < r3 < r4 < 1$;

When $\Delta C1 < C10$, $\Delta D1 < D10$ and $\Delta E1 < E10$, the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ is not adjusted;

When at least one of the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period is more than or equal to the standard value, adjust the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, specifically:

When $\Delta C1 \geq C10$, $\Delta D1 \geq D10$ and $\Delta E1 < E10$, or $\Delta C1 < C10$, $\Delta D1 \geq D10$ and $\Delta E1 < E10$, or $\Delta C1 < C10$, $\Delta D1 < D10$ and $\Delta E1 \geq E10$, select the first compensation coefficient $r1$ to compensate the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$,

12

and the compensated test period is $Ti*zm*yi*r1$ or $Ti*zm*zm*yi*r1$, which is taken as the second test period;

When $\Delta C1 \geq C10$, $\Delta D1 \geq D10$ and $\Delta E1 < E10$, or $\Delta C1 \geq C10$, $\Delta D1 < D10$ and $\Delta E1 \geq E10$, or $\Delta C1 < C10$, $\Delta D1 \geq D10$ and $\Delta E1 \geq E10$, select the second compensation coefficient $r2$ to compensate the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and the compensated test period is $Ti*zm*yi*r2$ or $Ti*zm*zm*yi*r2$, which is taken as the second test period;

When $\Delta C1 \geq C10$, $\Delta D1 \geq D10$ and $\Delta E1 \geq E10$, select the third compensation coefficient $r3$ to compensate the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and the compensated test period is $Ti*zm*yi*r3$ or $Ti*zm*zm*yi*r3$, which is taken as the second test period.

Specifically, set the standard values of multiple monitoring values according to the historical data of the thermal power equipment, judge that the thermal power equipment may have hidden dangers in the operation when a large difference exists between the monitoring value and the standard value in the first test period, set the compensation factor to shorten the test period, increase the number of tests, and monitor faults in a timelier manner for maintenance.

It can be seen that in the embodiment, the test period of the thermal power equipment is adjusted during the precision spot inspection by obtaining multi-dimensional data information of the thermal power equipment, which can effectively improve the effectiveness and accuracy of the test period.

In another preferred embodiment based on the above embodiment, as shown in FIG. 2, this embodiment provides a multidimensional precision spot inspection system for thermal power equipment, comprising:

A determining module, which is used to determine the equipment requiring precision spot inspection, mark the equipment as the equipment to be inspected, determine the importance index of the equipment to be inspected, and determine the initial test period of the equipment to be inspected according to the importance index;

An acquisition module, which is used to acquire the historical precision spot inspection information of the equipment to be inspected, adjust the initial test period of the equipment to be inspected according to the historical precision spot inspection information, and obtain the first test period;

An processing module, which is used to perform the precision spot inspection on the equipment to be inspected according to the first test period, obtain the precision spot inspection results within the first test period, continue performing the precision spot inspection on the equipment to be inspected according to the first test period when the precision spot inspection result is less than the standard value, adjust the first test period to obtain the second test period when the precision spot inspection result is more than or equal to the standard value, and perform the precision spot inspection on the equipment to be inspected according to the second test period.

Specifically, the determining module is also used to determine the initial test period of the equipment to be inspected according to the importance index, including:

Determine the importance index $A0$ of the equipment to be inspected, preset the first preset importance index $A1$, the second preset importance index $A2$, the third preset importance index $A3$ and the fourth preset importance index $A4$, and $A1 > A2 > A3 > A4$;

Preset the first preset test period $T1$, the second preset test period $T2$, the third preset test period $T3$ and the fourth preset test period $T4$, and $0 < T1 < T2 < T3 < T4$;

13

Determine the initial test period of the equipment to be inspected according to the relationship between the importance index A0 and each preset importance index:

When $A0 \geq A1$, select the first preset test period T1 as the initial test period of the equipment to be inspected;

When $A1 > A0 \geq A2$, select the second preset test period T2 as the initial test period of the equipment to be inspected;

When $A2 > A0 \geq A3$, select the third preset test period T3 as the initial test period of the equipment to be inspected;

When $A3 > A0 \geq A4$, select the fourth preset test period T4 as the initial test period of the equipment to be inspected;

After the i th preset test period T_i is selected as the initial test period of the equipment to be inspected, $i=1,2,3,4$, and then the initial test period of the equipment to be inspected is T_i at this time, which is denoted as the initial test period T_i .

It is understandable that in the above embodiment, the test period of the thermal power equipment is adjusted during the precision spot inspection by obtaining multi-dimensional data information of the thermal power equipment, which can effectively improve the effectiveness and accuracy of the test period.

The technicians in the field should understand that the embodiments of the application may be provided as methods, systems, or computer program products. Therefore, the application may take the form of a full hardware embodiment, full software embodiment, or combination of software and hardware embodiments. Further, the application may take the form of a computer program product implemented on one or more computer available storage media (including but not limited to disk memory, CD-ROM, optical memory, etc.) in which computer available program codes are contained.

The application is described by reference to the flowcharts and/or block diagrams of the methods, equipment (systems), and computer program products based on the embodiments of the application. It should be understood that each flow and/or block in the flowchart and/or block diagram, and a combination of flows and/or blocks in the flowchart and/or block diagram can be implemented by computer program instructions. The computer program instructions may be sent to a processor of a special-purpose computer, an embedded processor or other programmable data processing equipment to generate a machine, so that the instructions executed by the processor of the computer or other programmable data processing equipment generates a device used to implement functions specified in one or more flows of the flowchart and/or one or more blocks of the block diagram.

The computer program instructions may also be stored in a computer-readable memory capable of directing the computer or other programmable data processing equipment to work in a specific way, so that the instructions stored in the computer-readable memory generate manufactures including a command device, which implements functions specified in one or more flows of the flowchart and/or one or more blocks of the block diagram.

The computer program instructions may also be loaded onto the computer or other programmable data processing equipment, so that a series of operational steps are performed on the computer or other programmable data processing equipment to generate computer-implemented processing, and thus the instructions executed on the computer or other programmable equipment provide steps to implement functions specified in one or more flows of the flowchart and/or one or more blocks of the block diagram.

Finally, it should be noted that the above embodiments are used only to describe the technical proposal of the invention

14

and not to restrict it. Although the invention is described in detail by reference to the above embodiments, the ordinary technicians in the field should understand that the embodiments of the invention may still be modified or equivalently replaced, and any modification or equivalent replacement not deviating from spirit and scope of the invention should be covered by the claims of the invention.

The invention claimed is:

1. A multi-dimensional spot inspection system for thermal power equipment, comprising:

multiple testing instruments that are used for offline testing of thermal power equipment; one or more processors; memory storing instructions, executable by the one or more processors, that, when executed by the one or more processors, cause the one or more processors to:

read an equipment to be inspected, and an importance index of the equipment to be inspected, which are stored in the memory, and determine an initial test period of the equipment to be inspected according to the importance index;

acquire historical spot inspection information of the equipment to be inspected, which is stored in the memory, adjust the initial test period of the equipment to be inspected according to the historical spot inspection information, so as to obtain a first test period which will be stored in the memory;

send first testing instructions to the multiple testing instruments to perform the spot inspection on the equipment to be inspected according to the first test period, obtain the spot inspection results within the first test period, continue to send the first testing instructions to the multiple testing instruments to perform the spot inspection on the equipment to be inspected according to the first test period when the spot inspection result is less than the standard value, adjust the first test period to obtain a second test period when the spot inspection result is more than or equal to the standard value, and to send second testing instructions to the multiple testing instruments to perform the spot inspection on the equipment to be inspected according to the second test period; and

send maintenance instructions for the thermal power equipment to maintenance personnel based on the spot inspection results performed in the first test period or the spot inspection results performed in the second test period;

wherein when the one or more processors are used to determine the initial test period of the equipment to be inspected according to the importance index, the one or more processors are specifically used to:

determine the importance index A0 of the equipment to be inspected, preset the first preset importance index A1, the second preset importance index A2, the third preset importance index A3 and the fourth preset importance index A4, and $A1 > A2 > A3 > A4$;

preset the first preset test period T1, the second preset test period T2, the third preset test period T3 and the fourth preset test period T4, and $0 < T1 < T2 < T3 < T4$;

determine the initial test period of the equipment to be inspected according to the relationship between the importance index A0 and each preset importance index; when $A0 \geq A1$, select the first preset test period T1 as the initial test period of the equipment to be inspected;

when $A1 > A0 \geq A2$, select the second preset test period T2 as the initial test period of the equipment to be inspected;

15

when $A2 > A0 \geq A3$, select the third preset test period T3 as the initial test period of the equipment to be inspected; when $A3 > A2 \geq A4$, select the fourth preset test period T4 as the initial test period of the equipment to be inspected; after the i^{th} preset test period Ti is selected as the initial test period of the equipment to be inspected, $i=1,2,3,4$, and then the initial test period of the equipment to be inspected is Ti at this time, which is denoted as the initial test period Ti;

wherein when the one or more processors are used to acquire the historical spot inspection information of the equipment to be inspected, which is stored in the memory, adjust the initial test period of the equipment to be inspected according to the historical spot inspection information, so as to obtain the first test period, the one or more processors are specifically used to:

after determining the initial test period Ti, acquire the historical spot inspection information of the equipment to be inspected, which includes the historical spot inspection result data and the historical maintenance data of the equipment to be inspected, and adjust the initial test period Ti according to the historical spot inspection result data and the historical maintenance data; wherein,

after obtaining the historical inspection check information, determine whether the historical spot inspection result data and the historical maintenance data are obtained;

if only the historical spot inspection result data is obtained, determine the difference between two adjacent historical spot inspection result data, adjust the initial test period Ti according to the difference between two adjacent historical spot inspection result data to obtain the first test period;

if both the historical spot inspection result data and the historical maintenance data are obtained, determine the difference between two adjacent historical spot inspection result data, obtain the historical maintenance frequency of the equipment to be inspected, adjust the initial test period Ti according to the difference between two adjacent historical spot inspection result data, and then correct the adjusted initial test period Ti according to the historical maintenance frequency to obtain the first test period;

wherein when the one or more processors are used to adjust the initial test period Ti based on the difference between two adjacent historical spot inspection result data, the one or more processors are specifically used to:

after obtaining the historical spot inspection result data, establish the historical result matrix B according to the historical spot inspection result data, set B (C1, D1, E1, C2, D2, E2, C3, D3, E3, . . . , Cn, Dn, En), where C₁ to C_n are overall vibration values of the equipment to be inspected which are respectively obtained from the 1st spot inspection to the nth spot inspection, D₁ to D_n are infrared temperature values of the equipment to be inspected which are respectively obtained from the 1st spot inspection to the nth spot inspection, E₁ to E_n are ultrasonic values of the equipment to be inspected which are respectively obtained from the 1st spot inspection to the nth spot inspection;

preset the preset overall vibration difference c0, the preset standard infrared temperature difference d0 and the preset standard ultrasonic difference e0; preset the first preset adjustment coefficient z1, the second preset

16

adjustment coefficient z2 and the third preset adjustment coefficient z3, and $1 > z1 > z2 > z3 > 0.8$;

obtain the overall vibration values, the infrared temperature values and the ultrasonic values within the last two periods in the historical result matrix B, obtain two adjacent overall vibration differences, denoted as C_{n-1}-C_n, obtain two adjacent infrared temperature differences, denoted as D_{n-1}-D_n, and obtain two adjacent ultrasonic differences, denoted as E_{n-1}-E_n;

compare C_{n-1}-C_n, D_{n-1}-D_n and E_{n-1}-E_n with c0, d0 and e0 respectively, and select the adjustment coefficient according to the comparison results to adjust the initial test period Ti:

when C_{n-1}-C_n ≤ c0, D_{n-1}-D_n ≤ d0 and E_{n-1}-E_n ≤ e0, the initial test period Ti is not adjusted;

when C_{n-1}-C_n > c0, D_{n-1}-D_n ≤ d0 and E_{n-1}-E_n ≤ e0, or C_{n-1}-C_n ≤ c0, D_{n-1}-D_n > d0 and E_{n-1}-E_n ≤ e0, or C_{n-1}-C_n ≤ c0, D_{n-1}-D_n ≤ d0 and E_{n-1}-E_n > e0, select the first preset adjustment coefficient z1 to adjust the initial test period Ti, the adjusted initial test period is Ti*z1, and Ti*z1 is an integer;

when C_{n-1}-C_n > c0, D_{n-1}-D_n > d0 and E_{n-1}-E_n ≤ e0, or C_{n-1}-C_n ≤ c0, D_{n-1}-D_n > d0 and E_{n-1}-E_n > e0, or C_{n-1}-C_n > c0, D_{n-1}-D_n ≤ d0 and E_{n-1}-E_n > e0, select the second preset adjustment coefficient z2 to adjust the initial test period Ti, the adjusted initial test period is Ti*z2, and Ti*z2 is an integer;

when C_{n-1}-C_n > c0, D_{n-1}-D_n > d0 and E_{n-1}-E_n > e0, select the third preset adjustment coefficient z3 to adjust the initial test period Ti, the adjusted initial test period is Ti*z3, and Ti*z3 is an integer;

after the mth preset adjustment coefficient zm is selected to adjust the initial test period Ti, m=1,2,3, and then the adjusted initial test period Ti*zm is taken as the first test period.

2. The multi-dimensional spot inspection system for thermal power equipment of claim 1, wherein when the adjusted initial test period Ti*zm is taken as the first test period, the one or more processors are specifically used to:

after establishing the historical result matrix B based on the historical spot inspection result data, determine the overall vibration average c01, the infrared temperature average d01 and the ultrasonic average e01 according to the historical result matrix B, where $c01 = (C_1 + C_2 + C_3 + \dots + C_n)/n$, $d01 = (D_1 + D_2 + D_3 + \dots + D_n)/n$, and $e01 = (E_1 + E_2 + E_3 + \dots + E_n)/n$;

preset the preset standard overall vibration average ca, the preset standard infrared temperature average da and the preset standard ultrasonic average ea, and adjust the adjusted initial test period Ti*zm again according to the relationship between the overall vibration average c01, the infrared temperature average d01 and the ultrasonic average e01, and the preset standard overall vibration average ca, the preset standard infrared temperature average da and the preset standard ultrasonic average ea respectively:

when c01 ≤ ca, d01 ≤ da and e01 ≤ ea, the adjusted initial test period Ti*zm is not adjusted again;

when c01 > ca, d01 ≤ da and e01 ≤ ea, or c01 ≤ ca, d01 > da and e01 ≤ ea, or c01 ≤ ca, d01 ≤ da and e01 > ea, select the first preset adjustment coefficient z1 to adjust the adjusted initial test period Ti*zm again, the readjusted test period is Ti*zm*z1, and Ti*zm*z1 is an integer;

when c01 > ca, d01 > da and e01 ≤ ea, or c01 ≤ ca, d01 > da and e01 ≤ ea, or c01 > ca, d01 ≤ da and e01 > ea, select the second preset adjustment coefficient z2 to adjust the

17

adjusted initial test period $Ti*zm$ again, the readjusted test period is $Ti*zm*z2$, and $Ti*zm*z2$ is an integer; when $c01>ca$, $d01>da$ and $e01>ea$, select the third preset adjustment coefficient $z3$ to adjust the adjusted initial test period $Ti*zm$ again, the readjusted test period is $Ti*zm*z3$, and $Ti*zm*z3$ is an integer; after the m^{th} preset adjustment coefficient zm is selected to adjust the adjusted initial test period $Ti*zm$ again, $m=1,2,3$, and then the adjusted initial test period $Ti*zm*zm$ is taken as the first test period.

3. The multi-dimensional spot inspection system for thermal power equipment of claim 2, wherein when adjusting the initial test period Ti according to the difference between two adjacent historical spot inspection result data and then correcting the adjusted initial test period Ti according to the historical maintenance frequency to obtain the first test period, the one or more processors are specifically used to:

after obtaining the historical maintenance data, determine the historical maintenance frequency $Q0$ of the equipment to be inspected according to the historical maintenance data;

preset the first preset maintenance frequency $Q1$, the second preset maintenance frequency $Q2$, the third preset maintenance frequency $Q3$ and the fourth preset maintenance frequency $Q4$, and $Q1<Q2<Q3<Q4$; preset the first preset correction coefficient $y1$, the second preset correction coefficient $y2$, the third preset correction coefficient $y3$ and the fourth preset correction coefficient $y4$, and $1<y1<y2<y3<y4<1.5$;

after obtaining the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, select the correction coefficient according to the relationship between the historical maintenance frequency $Q0$ and each preset maintenance frequency, and correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$:

when $Q1\leq Q0<Q2$, select the first preset correction coefficient $y1$ to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, and the corrected test period is $Ti*zm*y1$ or $Ti*zm*zm*y1$;

when $Q2\leq Q0<Q3$, select the second preset correction coefficient $y2$ to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, and the corrected test period is $Ti*zm*y2$ or $Ti*zm*zm*y2$;

when $Q3\leq Q0<Q4$, select the third preset correction coefficient $y3$ to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, and the corrected test period is $Ti*zm*y3$ or $Ti*zm*zm*y3$;

when $Q4\leq Q0$, select the fourth preset correction coefficient $y4$ to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, and the corrected test period is $Ti*zm*y4$ or $Ti*zm*zm*y4$;

after selecting the i^{th} preset correction coefficient yi to correct the adjusted initial test period $Ti*zm$ or $Ti*zm*zm$, $i=1,2,3,4$, and the corrected test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ is taken as the first test period.

4. The multi-dimensional spot inspection system for thermal power equipment of claim 3, wherein when the corrected test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ is taken as the first test period, the one or more processors are specifically used to:

18

perform the spot inspection on the equipment to be inspected according to the first test period, and obtain the spot inspection results within the first test period, which includes the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period; wherein,

when the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period are less than the standard values, do not adjust the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and continue to perform the spot inspection on the equipment to be inspected according to the first test period;

when at least one of the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period is more than or equal to the standard value, adjust the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ to obtain the second test period, and perform the spot inspection on the equipment to be inspected according to the second test period.

5. The multi-dimensional spot inspection system for thermal power equipment of claim 4, wherein when presetting the standard values which include the preset standard overall vibration value $C10$, the preset standard infrared temperature value $D10$ and the preset standard ultrasonic value $E10$, and obtaining the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period, the one or more processors are specifically used to:

preset the first compensation coefficient $r1$, the second compensation coefficient $r2$, the third compensation coefficient $r3$ and the fourth compensation coefficient $r4$, and $0.8<r1<r2<r3<r4<1$;

when $\Delta C1<C10$, $\Delta D1<D10$ and $\Delta E1<E10$, the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$ is not adjusted;

when at least one of the overall vibration value $\Delta C1$, the infrared temperature value $\Delta D1$ and the ultrasonic value $\Delta E1$ within the first test period is more than or equal to the standard value, adjust the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, specifically:

when $\Delta C1\geq C10$, $\Delta D1<D10$ and $\Delta E1<E10$, or $\Delta C1<C10$, $\Delta D1\geq D10$ and $\Delta E1<E10$, or $\Delta C1<C10$, $\Delta D1<D10$ and $\Delta E1\geq E10$, select the first compensation coefficient $r1$ to compensate the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and the compensated test period is $Ti*zm*yi*r1$ or $Ti*zm*zm*yi*r1$, which is taken as the second test period;

when $\Delta C1\geq C10$, $\Delta D1\geq D10$ and $\Delta E1<E10$, or $\Delta C1\geq C10$, $\Delta D1<D10$ and $\Delta E1\geq E10$, or $\Delta C1<C10$, $\Delta D1\geq D10$ and $\Delta E1\geq E10$, select the second compensation coefficient $r2$ to compensate the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and the compensated test period is $Ti*zm*yi*r2$ or $Ti*zm*zm*yi*r2$, which is taken as the second test period;

when $\Delta C1\geq C10$, $\Delta D1\geq D10$ and $\Delta E1\geq E10$, select the third compensation coefficient $r3$ to compensate the first test period $Ti*zm*yi$ or $Ti*zm*zm*yi$, and the compensated test period is $Ti*zm*yi*r3$ or $Ti*zm*zm*yi*r3$, which is taken as the second test period.

* * * * *